INCENDIOS_PY · SOVEREIGN GEOAI SYSTEM

Decoupled Biophysical Ignition Hazard & Tactical Wildfire Propagation Framework

DAILY HAZARD NOWCAST · 1-km² RESOLUTION

PHASE 1 Multisource Ingestion

Paraguay (1 km² Grid)
VIIRS Fire Hotspots



NASA FIRMS (VIIRS 375m)

Suomi-NPP, NOAA-20, NOAA-21
Bow-Tie filtered (scan ≤ 1.5 km)

ERA5-Land Reanalysis

0.25° grid (640 national nodes)
VPD, Wind, CDD, FFMC, ISI

Fuel Cover (MapBiomas)

Annual C2, prior year
400.006 cells

STAGE 1: KINEMATIC BIO-PHYSICAL GATING

Condition: VPD ≥ 0.8 kPa AND Rainfall 48h < 10 mm AND Fuel = 1

- Automatically excludes water, wet savannas & saturated air cells.
- Discards 39.2% of non-fire national cell-days (89.94% fire retention).
- Focuses compute from 1,168.8M cell-days to high-risk candidates.

Physical Drivers

Thermal Foci (t-1)

PHASE 2 Decoupled Dual-Layer GeoAI Engine

MODE 1: MULTISCALE SOCIO-ECOLOGICAL HYBRID (M3)

Features: Weather + Fuel Structure + Anthropogenic Priors (10/25km)

- STRICT CAUSAL ISOLATION: No fire history lags allowed (t-1 = 0).
- Drivers: VPD max, Thermal amplitude, Wind speed, Pasture fire pressure.
- Calibrated via Monotonic Platt Scaling & Split Conformal Prediction.

Output: P(Ignition M3) --> Pre-suppression & Forest Patrol

MODE 2: TACTICAL FIRE SPREAD & CONTAGION

Features: Physical Drivers + Spatial Proximity Kernel (t-1)

- Causal spatial memory: dist_foco_t1_km, focos_25km_t1, frp_25km_t1.
- Models convective advection & active flame front advancement.
- Evaluated with 20-km Dead-Zone spatial buffer to prevent leakage.

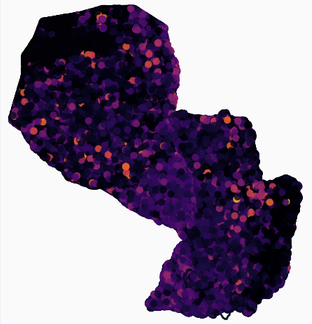
Output: P(Propagation) --> Tactical Firefighting Deployment

P(Ignition) Forecast

P(Spread) Validation

PHASE 3 Sovereign Alert & Bench

Operational Alert Map
(1 km² Resolution)



OPERATIONAL METRICS

• Pipeline Latency:	2.46 s
• Sovereign Grid:	400,006 cells
• Population Precision (Propagation):	9.8 %
• Chaco Spread ROC:	0.903
• Oriental Hybrid ROC:	0.754
• Chaco Hybrid ROC:	0.802

Gran Chaco Benchmarking on Blind Out-of-Time Test Set (2024–2025)

